

some photons representing state 0 and others representing state 1, and Robin wishes to break the code, he must measure the states of the photons. This measurement alters the states, alerting both Bimal and Rahim to the intrusion.

In quantum entanglement, two photons are linked in such a way that if the state of one photon changes, the other changes instantly. Therefore, if Robin attempts to intercept Bimal's message, it becomes impossible in this scenario. These principles were experimentally verified in 1992, but widespread application has taken longer to materialise.

China has made notable progress in this field. In 2013, a group of researchers announced that they had used quantum key distribution on a hot-air balloon carrying a payload 20km above a ground station. This is considered the world's first experimental quantum satellite. Over the last decade, China has advanced significantly and currently operates three satellites and four ground stations, forming the world's largest QKD network.

Where does India stand? In October last year, researchers at the Raman Research Institute in Bengaluru reported that the Indian Astronomical Observatory in Hanle, Ladakh, possesses ideal infrastructure to support satellites equipped for QKD. They also claim that the signal loss at this site will be six decibels lower than in China. Recently, India's space research organisation, ISRO, established a quantum communication system over a distance of 300m through a vacuum medium.

The United Nations recently declared 2025 the International Year of Quantum (IYQ) Science and Technology, to increase public awareness of quantum science and technology. In an era shaped by artificial intelligence, one thing is clear: progress is difficult for a developing nation without access to advanced computing capabilities. For this reason, the Ministry of Science and Technology of the government of India launched the 'National Quantum Mission' (NQM) in April 2023, with a budget of ₹60.03 billion. Its goal is to make the country proficient in quantum computing and quantum information—essentially enabling a quantum revolution—by 2031.

The NQM has developed four thematic hubs (T-Hubs) in key technological verticals, including quantum computing, communication, sensing, metrology, materials, and devices. The 'Quantum Satellite' is part of this mission, and the NQM is progressing toward establishing a robust satellite-based quantum communication network spanning up to 2000km within and beyond the country. **EFY**

SMART E-HORN DRIVER with Diotec MOSFET

Modern electronic horns (E-Horns) replace mechanical horn mechanisms with solid-state tone generation and MOSFET-based power control. Mechanical horns are low cost but typically fail within two years due to coil burnout and loss of resonance tuning. By using a MOSFET driving stage, the excitation frequency and amplitude are precisely controlled, ensuring operation at the transducer's resonant frequency. This improves efficiency, reduces mechanical stress, and extends horn lifetime. This application note presents a compact microcontroller-based E-Horn solution using a Diotec MOSFET that provides reverse-polarity protection, low conduction loss, and reliable high-current switching



DI101N10PQ this MOSFET used for the Switching Purpose:

- 100 V, 101 A N-Channel MOSFET
- Low RDS(on) for reduced conduction loss
- High avalanche rating for inductive load switching
- Suitable for 12 V & 24 V automotive horn applications

DI060N06PQ this MOSFET used for the Reverse Polarity Protection:

- 60 V, 180 A N-Channel MOSFET
- Ultra-low RDS(on) for minimal heating
- High SOA performance for repetitive switching
- Ideal for PWM-controlled horn drivers

Use of these MOSFETs for E-Horn Design

MOSFETs DI101N10PQ and DI060N06PQ are tailored for inductive load switching and provide:

- Robust electrical stress handling
- High peak current capability
- Low conduction and switching losses
- Improved thermal efficiency in compact designs

FOR MORE INFO



Diotec Electronics India Private Limited

Survey No.614, Hissa No.3 to 9, Saswad Village, Purandar Taluka, Dist: Pune, Maharashtra - 412 301

Tel: +91-(0)-2115-225047/222523

Email: info@diotec.in; <http://www.diotec.com/>